Pontryagin-Based Solver with Universally Smoothed Hamiltonian, Adaptive ΔT, and PA-Bundle Refinement

March 15, 2026

Presentation Overview

Previous Meeting

To do:

- Plot a horizontal line at the minimum value $K\sqrt{\Delta t_{\text{max}}}$ to verify that ρ_n is properly constrained.
- Set the constant K as the average of the ρ_n's to keep refinement consistent under problem scaling.
- Defining ρ_n as

$$\bar{\rho}_n := \mathrm{sgn}(\rho_n)\max(|\rho_n|, K\sqrt{\Delta t_{\max}}). \quad (1)$$

  can bypass the floor as for numerical zero value for ρ_n, sgn$(\rho_n) = 0$.
- Check Current PA-Bundle Hamiltonian Surrogate initialization:

The Followign example is taken from Karlsson et. al. [1]:

Hypersensitive optimal control

Minimize

$$\int_0^{25} \left(X(t)^2 + \alpha(t)^2\right)\, \mathrm{d}t + \gamma\left(X(25) - 1\right)^2 \tag{2}$$

subject to

$$X'(t) = -X(t)^3 + \alpha(t), \qquad 0 < t \leq 25, \tag{3}$$

$$X(0) = 1. \tag{4}$$

for $\gamma = 1e6$. The Hamiltonian is then given by

$$H(x, \lambda) = \min_{\alpha}\{-\lambda x^3 + \lambda\alpha + x^2 + \alpha^2\} = -\lambda x^3 - \frac{\lambda^2}{4} + x^2 \tag{5}$$

Inital Issues: H and $\bar{H}$

PA-Bundle surrogate $\bar{H}$

- The PA bundle stores a finite set of controls

$$\mathcal{U}_{\text{bundle}} = \{u^{(1)}, \dots, u^{(K)}\} \subset A.$$

For each stored control $u^{(i)}$, define the affine function of p

$$\phi_i(p; x, t) := g_i(x,t)^\top p + d_i(x,t), \qquad g_i(x,t) := f(x, u^{(i)}, t), \quad d_i(x,t) := \ell(x, u^{(i)}, t).$$

$$\bar{H}(p, x, t) := \min_{i=1,\dots,K} \phi_i(p; x, t) = \min_{u \in \mathcal{U}_{\text{bundle}}} \mathcal{H}(p, x, u, t).$$

Hamiltonian H

- The Hamiltonian builds a finite candidate set

$$\mathcal{U}_{\text{cand}} \subset A \quad \text{(e.g., box corners + bundle controls)} \tag{6}$$

$$H(p, x, t) = \min_{\mathcal{U}_{\text{cand}}} (p^\top f(x, u, t) + \ell(x, u, t) \tag{7}$$

Changes

Before

- Initialize bundle with a *fixed* small set of controls:
$$U^{(0)}_{\text{bundle}} = \{u_{\min},\, u_{\text{mid}},\, u_{\max}\} \tag{8}$$
(or $\{0\}$ if no bounds). (9)

- Surrogate Hamiltonian (used by Newton via smoothing):
$$\bar{H}(p,x,t) = \min_{u \in U_{\text{bundle}}} h_u(p;x,t), \tag{10}$$
$$h_u(p;x,t) = p^\top f(x,u,t) + \ell(x,u,t). \tag{11}$$

- Limitation: $U^{(0)}_{\text{bundle}}$ may miss the *active controls* on the coarse trajectory.

Now (trajectory-based)

1. **Coarse solve** on initial mesh to obtain $(X^{(0)}, P^{(0)})$.
2. Select k support nodes
$$\{(x_i,p_i,t_i)\}_{i\in\mathcal{I}}, \qquad |\mathcal{I}| = k. \tag{12}$$
3. **Approximate active control** at each support node via cheap search over candidates $\mathcal{G} \subset A$ (e.g., 1D grid):
$$\hat{u}_i \in \arg\min_{u\in\mathcal{G}} h_u(p_i;x_i,t_i). \tag{13}$$
4. **Enrich bundle**:
$$U^{(0)}_{\text{bundle}} \leftarrow U^{(0)}_{\text{bundle}} \cup \{\hat{u}_i :\ i \in \mathcal{I}\}. \tag{14}$$
5. **Re-solve once** with the enriched bundle (same mesh, same δ).

Goal: start Newton with a surrogate $\bar{H}$ that already contains controls relevant to the visited region of (p,x,t).

Changes

Before (bundle-restricted candidates)

- Numerical evaluation of the Hamiltonian used (almost) the same controls already stored in the bundle:

$$H_{\text{approx}}(p,x,t) := \min_{u\in U_{\text{cand}}} h_u(p;x,t), \tag{15}$$

$$U_{\text{cand}} \approx U_{\text{bundle}} \cup \{\text{bounds extremes}\}. \tag{16}$$

- Surrogate (bundle) remains

$$\bar{H}(p,x,t) := \min_{u\in U_{\text{bundle}}} h_u(p;x,t). \tag{17}$$

- Consequence on nodes (P_n, X_n, t_n):

$$H_{\text{approx}}(P_n,X_n,t_n) \approx \bar{H}(P_n,X_n,t_n) \tag{18}$$

$$\Rightarrow\ \bar{H} - H_{\text{approx}} \approx 0. \tag{19}$$

Now (added 1D grid search for H)

- Keep $\bar{H}$ unchanged (still bundle-only):

$$\bar{H}(p,x,t) := \min_{u\in U_{\text{bundle}}} h_u(p;x,t). \tag{20}$$

- Make H *strictly richer* by enlarging the candidate set:

$$H_{\text{approx}}(p,x,t) := \min_{u\in U^+_{\text{cand}}} h_u(p;x,t), \tag{21}$$

$$U^+_{\text{cand}} := U_{\text{bundle}} \cup \{\text{bounds extremes}\} \cup \mathcal{G}, \tag{22}$$

where $\mathcal{G} \subset [u_{\min}, u_{\max}]$ is a cheap 1D grid.

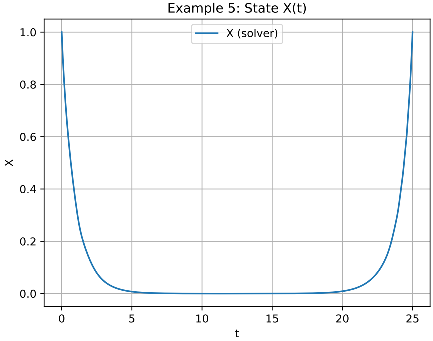
- Expected inequality (pointwise):

$$H_{\text{approx}}(p,x,t) \le \bar{H}(p,x,t) \tag{23}$$

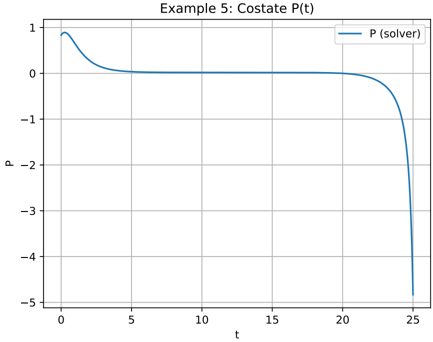
$$\Rightarrow\ \text{gap}(p,x,t) := \bar{H} - H_{\text{approx}} \ge 0. \tag{24}$$

- Practical effect: η_{PA} becomes informative and triggers adaptive plane enrichment when needed.

We consider the same example with $s = 0.25$ and $M = 2$ (since $c \approx 1$).



(a) Solution $\bar{X}$ of the discrete optimization problem (2) for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{\text{PA}} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$



(b) Solution $\bar{\lambda}$ of the discrete optimization problem (2) for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{\text{PA}} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$

Numerical Results

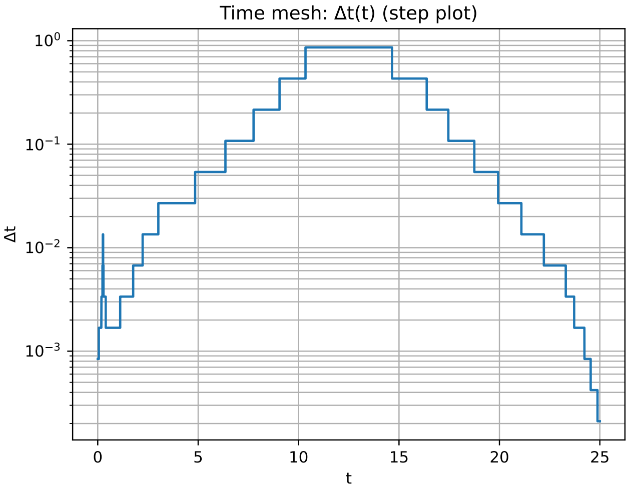
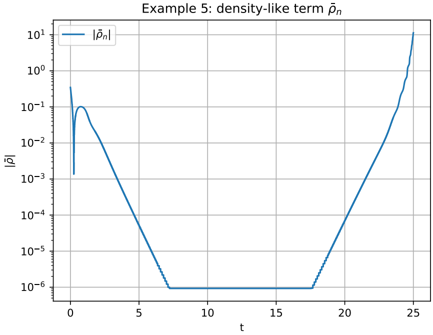
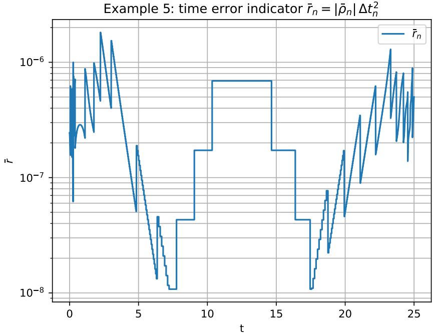


Figure: Mesh Δt for the optimization problem (2) for $\varepsilon_{\text{time}} = 1e - 2$, $\varepsilon_{\text{PA}} = 1e - 2$, $\varepsilon_{\delta} = 1e - 2$, $\delta_0 = 0.02$

Numerical Results



(a) Error densities, $|\bar{\rho}|$ for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{\text{PA}} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$



(b) Error indicators $\bar{r}_n$ of the discrete optimization problem (2) for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{\text{PA}} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$

Numerical Results

Table: Initial Parameters

N	M	δ_0	$\varepsilon_{\text{time}}$	ε_{PA}	ε_{δ}
30	3	0.02	1e-2	1e-2	1e-2

Table: Numerical Results

MaxIter	N	M	δ	η_{time}	η_{PA}	η_{δ}	Newton Residual
25	3558	18	0.01	1.82e-6	0.0021	0.1678	7.026e-9

Questions

(i) Changes made in the solver are example oriented, more general changes can be proposed, for instance:

- Optimal candidates control allocation for the Hamiltonian for dimension $m > 1$.
- PA surrogate initialization should be based in the nature of the problem.

References

Karlsson, J., Larsson, S., Sandberg, M., Szepessy, A., Tempone, Raul
An error estimate for symplectic euler approximation of optimal control problems.
arxiv, (2014).